

Credit Card Fraud Detection

Final Results

Models: XGBoost · Isolation Forest · Autoencoder · Ensemble

This report summarises detection performance on the held-out test split.

Metrics shown match the dashboard exactly.

Dataset Summary

Field	Value
Dataset	Credit Card
Total Samples	283,726
Normal Samples	283,253
Fraud Samples	473
Fraud Ratio	0.1667%
Number of Features	31
Test Samples	57,124
Test Fraud	473
Contamination	0.001667

Hyperparameters

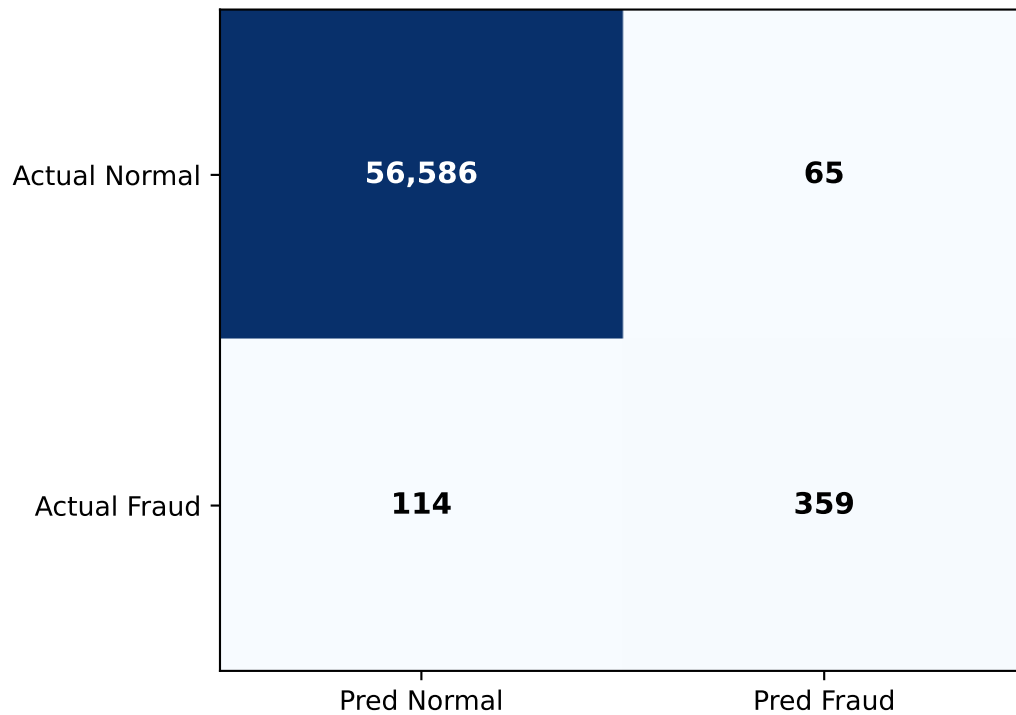
Model	Hyperparameters
XGBoost	n_estimators=300, max_depth=6, learning_rate=0.1, scale_pos_weight=576 eval_metric='logloss', random_state=42
Isolation Forest	n_estimators=300, max_samples=0.8, contamination=0.001667 random_state=42
Autoencoder	encoding_dim=14, epochs=20, batch_size=2048 threshold=percentile(1-contamination)
Ensemble	Mean of min-max normalized scores: XGBoost + Isolation Forest + Autoencoder. Threshold at training-time percentile.

Per-Model Evaluation

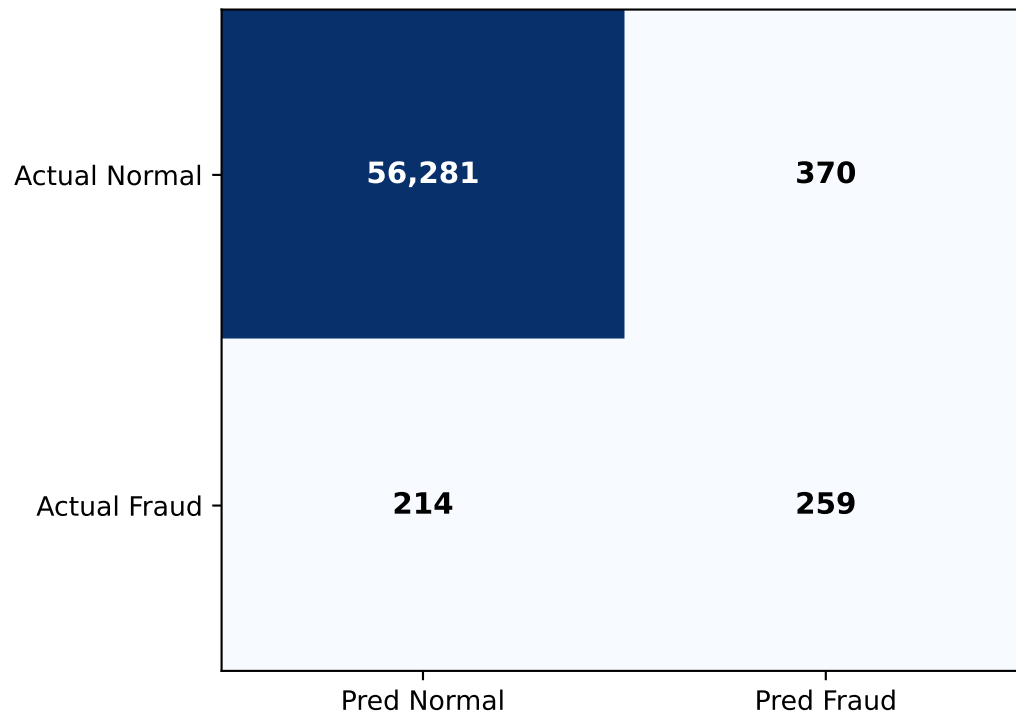
Rank	Model	Precision	Recall	F1-score	ROC-AUC	PR-AUC	TN	FP	FN	TP
1	XGBoost	0.8470	0.7580	0.8000	0.9620	0.8140	56586	65	114	359
2	Isolation Forest	0.4120	0.5470	0.4700	0.8930	0.3510	56281	370	214	259
3	Autoencoder	0.3670	0.4840	0.4175	0.8710	0.2980	56256	395	244	229
4	Ensemble	0.9230	0.2530	0.3971	0.9480	0.8060	56641	10	353	120

Confusion Matrices — Credit Card

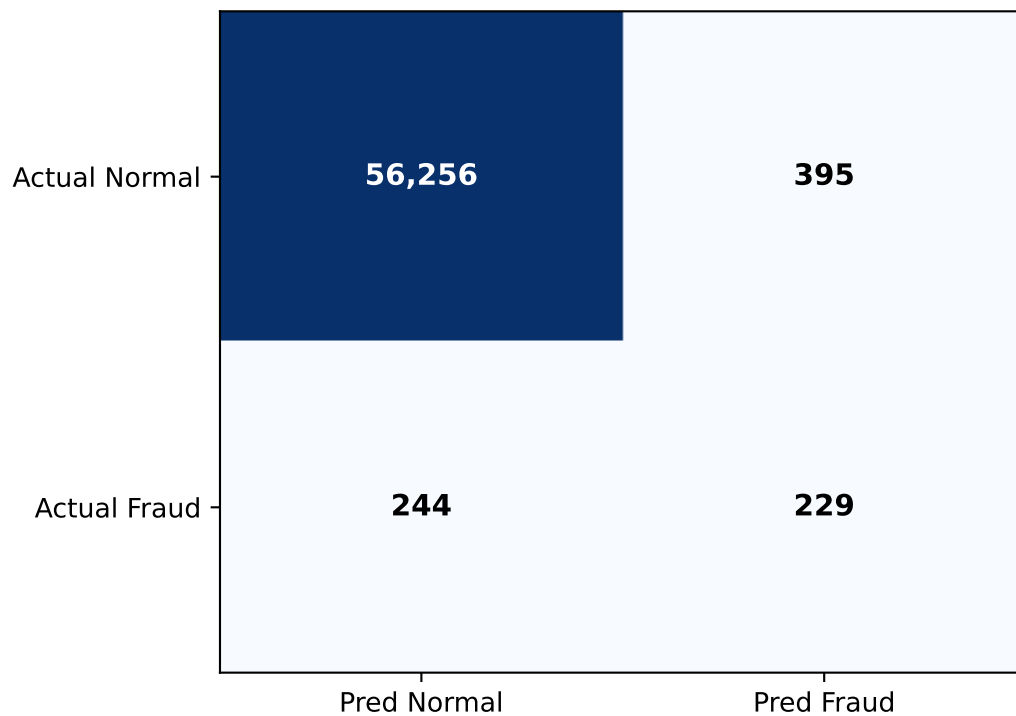
XGBoost



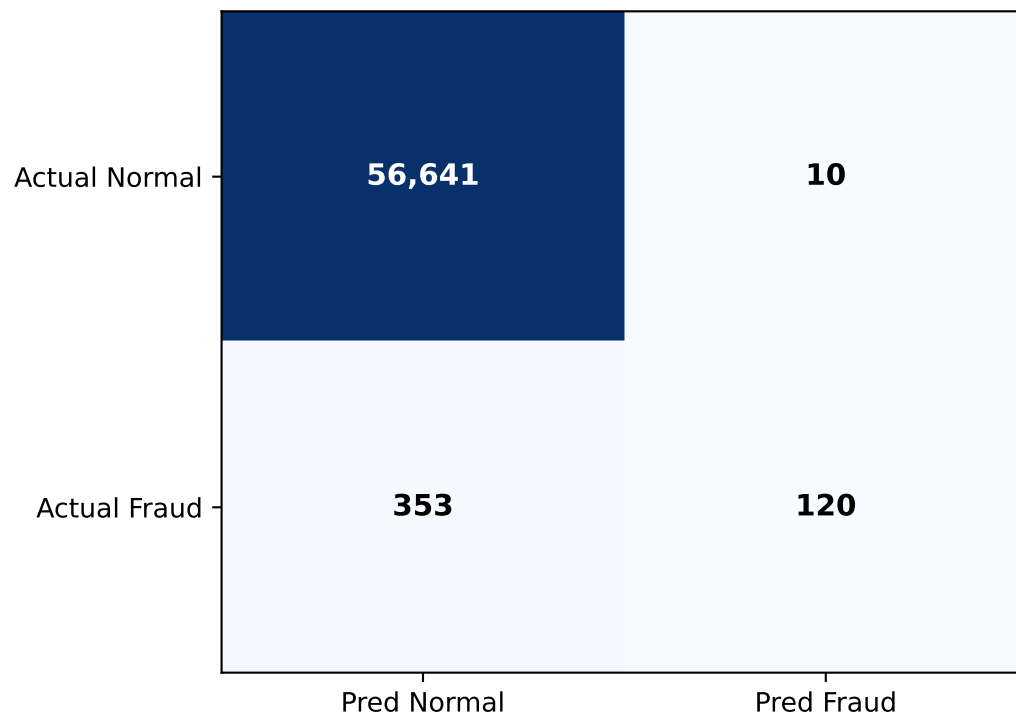
Isolation Forest



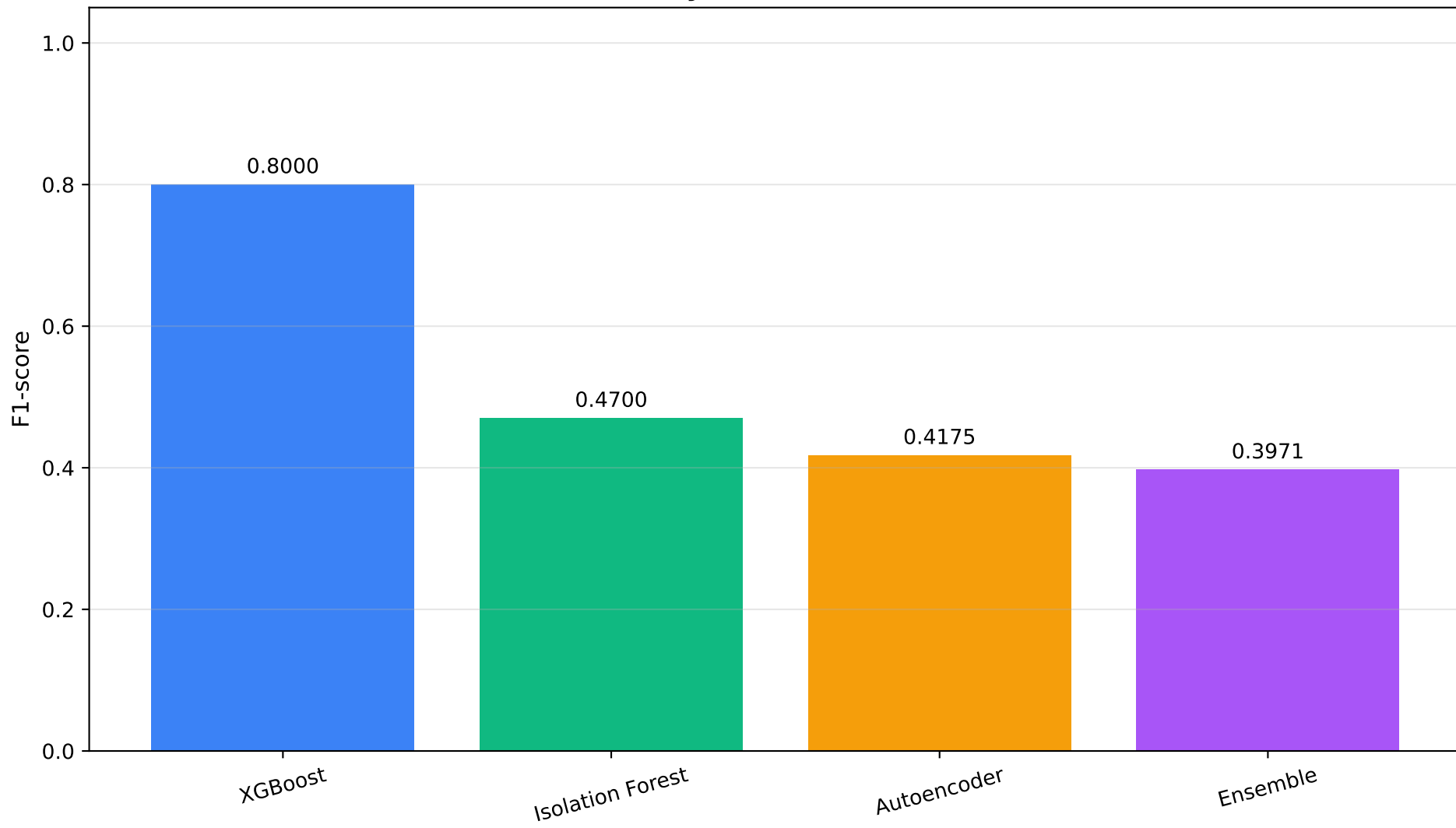
Autoencoder



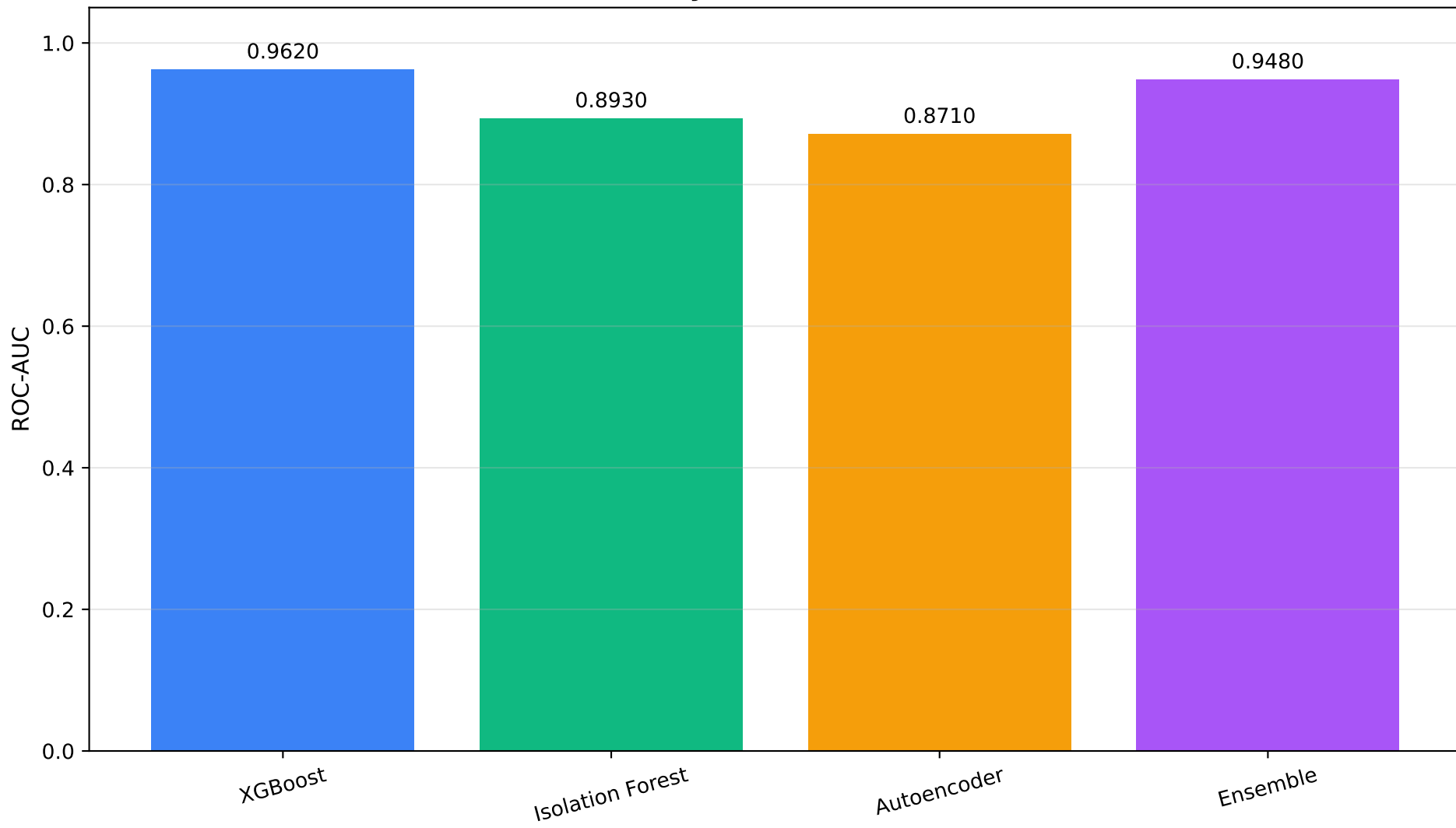
Ensemble



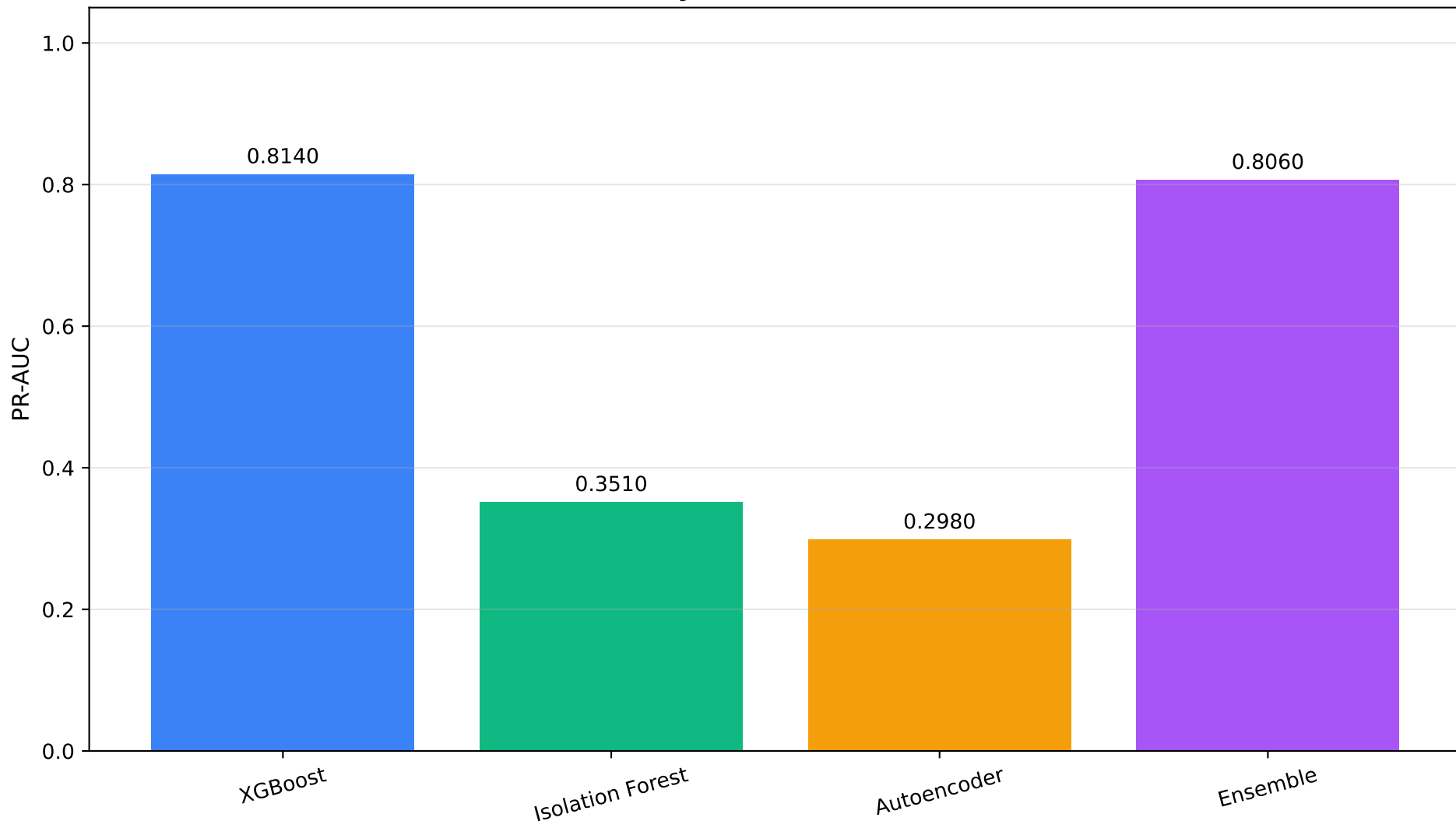
F1-score by Model — Credit Card



ROC-AUC by Model — Credit Card



PR-AUC by Model — Credit Card



Precision vs Recall — Credit Card

